EDUAI ACADEMIC SMART NOTES

Generated: October 29, 2026

University Curriculum Series

Introduction to Machine Learning

Section Topic: Model Evaluation, Cross-Validation & Diagnostics

Executive Summary: Covers rigorous experimental methodology, stratified cross-validation, leak-free pipeline engineering, and hyperparameter optimization.

1. Core Learning Objectives

- Design leak-free evaluation schemes using K-Fold and Stratified Cross-Validation
- Diagnose model health through Learning Curves and Validation Curves
- Apply Hyperparameter Tuning strategies: Grid Search, Random Search, and Bayesian Optimization
- Implement calibration curves and Brier scores for probability reliability

2. Conceptual Breakdown & Technical Formulations

1. The Generalization Principle

The ultimate test of any machine learning model is its generalization capability on independent, identically distributed (i.i.d.) unseen data. Splitting a dataset arbitrarily into training and testing sets is susceptible to statistical variance and data leakage.

2. Cross-Validation Paradigms

- **Hold-out Method**: Simple 70/15/15 train/val/test split. Fast, but vulnerable on small datasets.
- **K-Fold Cross-Validation**: Divides dataset into K disjoint partitions. Trains on K-1 folds and tests on the remaining fold, rotating K times. The overall performance is the sample average:

\text{CV}_K = \frac{1}{K} \sum_{k=1}^K \text{Metric}_k

- **Stratified K-Fold**: Preserves the exact class proportion across every fold. Critical for imbalanced datasets.
- **Time-Series Split**: Forward-chaining split that prevents temporal lookahead bias.

3. Learning Curves Diagnostic Framework

Plotting training score vs validation score as sample size N increases diagnoses capacity issues:

- **High Bias (Underfitting)**: Both training and validation errors plateau at an unacceptably high error rate. Adding more data will not help; a more expressive model or richer features are required.
- **High Variance (Overfitting)**: Training error is near zero, but validation error remains significantly higher. Adding training samples or applying regularization narrows the generalization gap.

4. Hyperparameter Optimization

$$\theta^* = \arg\min_{\theta \in \Theta} \text{CV}(\mathcal{M}_\theta)$$

1. **Grid Search**: Exhaustive evaluation over a discrete Cartesian product grid. Computationally expensive (O(S^P)).

2. **Randomized Search**: Samples candidate configurations randomly from probability distributions. Much faster and empirically effective.

3. **Bayesian Optimization**: Fits a Gaussian Process surrogate model to map hyperparameter spaces to expected performance improvements.

5. Python Pipeline Implementation

```
from sklearn.model_selection import StratifiedKFold, GridSearchCV
from sklearn.svm import SVC
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
pipe = Pipeline([
('scaler', StandardScaler()),
('svc', SVC(kernel='rbf'))
])
param_grid = {
'svc__C': [0.1, 1.0, 10.0],
'svc__gamma': ['scale', 'auto', 0.01]
}
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
grid = GridSearchCV(pipe, param_grid, cv=cv, scoring='roc_auc', n_jobs=-1)
grid.fit(X_train, y_train)
print(f"Optimal Hyperparameters: {grid.best_params_}")
print(f"Best Stratified ROC-AUC: {grid.best_score_:.4f}")
```

6. Exam-Important Concepts

- **Data Leakage in Preprocessing**: Performing feature scaling or imputation on the entire dataset prior to splitting leaks validation statistics into the model. Always execute transformations inside an estimator pipeline!

3. Self-Assessment & Exam Checklist

Topic Milestone	Mastery Check
Can you formulate the primary mathematical loss / model objective?	[ ] Confirmed
Can you distinguish parametric vs non-parametric assumptions?	[ ] Confirmed
Are you prepared to calculate precision, recall, and F1 metrics?	[ ] Confirmed
Do you understand the difference between L1 sparsity and L2 shrinkage?	[ ] Confirmed